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$$\int \frac{2x^2 - 5x + 7}{x^3 - 7x + 6} dx$$

$$x^3 - 7x + 6 = 0 \Leftrightarrow (x - 1)(x - 2)(x + 3) = 0$$

$$\Rightarrow \frac{2x^2 - 5x + 7}{(x - 1)(x - 2)(x + 3)} = \frac{A}{(x - 1)} + \frac{B}{(x - 2)} + \frac{C}{(x + 3)}$$

$$A = \left( \frac{2x^2 - 5x + 7}{(x - 2)(x + 3)} \right) \Big|_{x=1} = -1$$

$$B = \left( \frac{2x^2 - 5x + 7}{(x - 1)(x + 3)} \right) \Big|_{x=2} = 1$$

$$C = \left( \frac{2x^2 - 5x + 7}{(x - 1)(x - 2)} \right) \Big|_{x=-3} = 2$$

$$\Rightarrow \int \frac{2x^2 - 5x + 7}{x^3 - 7x + 6} dx = - \int \frac{1}{x - 1} dx + \int \frac{1}{x - 2} dx + 2 \int \frac{1}{x + 3} dx$$

$$= -\ln|x - 1| + \ln|x - 2| + 2\ln|x + 3| + c$$

## References